



CS & IT ENGINEERING



Algorithms

Divide & Conquer

Lecture No.- 08



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Masters Method

Topics to be Covered



Topic

Topic

Topic

P48s on Master's Mtd

Long Integer Multiplication



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



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TTTe

hw1: $T(n) = 6T(n/3) + n^2 \log n$

$$\left. \begin{array}{l} a=6 \\ b=3 \\ f(n)=n^2 \log n \end{array} \right\} \checkmark$$

$$\log_b a = \log_3 6 = \underline{\underline{1.63}}$$

QW2:-

$$T(n) = 4T(n/2) + \log n$$

$$a = 4$$

$$b = 2$$

$$F(n) = \log n$$

} ✓

$$\log_b a = \log_2 4 = \underline{\underline{2}}$$

Case:-

Is $\log n = O(n^{2-\epsilon})$, some $\epsilon > 0$?

True

$$T(n) = O(n^2)$$

p48

$$T(n) = n, \quad n \leq 3$$

80.7%

$$T(n) = T(n/3) + c \cdot n, \quad \text{otherwise}$$

TC=?

A) $\Theta(n)$

(c) $\Theta(n^2)$

B) $\Theta(n \log n)$

D) $\Theta(n^2 \log n)$

Case 1 : Is $c_n = O(n^{0-\epsilon})$, some $\epsilon > 0$?

→ Fails

↓
dec function

Case 2:-

$$\text{Is } f(n) = O(n^{\log b^a} * (\log n)^k)$$

a) $k \geq 0$?
b) $k = -1$? } No

$$Cn = O(n^0 * (\log n)^k)$$

$$Cn = O((\log n)^k)$$

Fails

Cons:- $f(n) = \Omega(n^{0+\epsilon}), \epsilon > 0?$
 $C \times n = \Omega(n^\epsilon), \text{ some } \epsilon > 0?$
 $\epsilon = 0.1$

and

True
 $a * f(n/b) \leq \delta * f(n)$

$1 * C(n/3) \leq \delta * C * n$

$\frac{Cn}{3} \leq \delta * Cn \Rightarrow \left(\frac{1}{3} \leq \delta\right) \quad \delta < 1 \quad \underline{\underline{\text{True}}}$

$T(n) = \Theta(f(n)) = \underline{\underline{\Theta(Cn)}} = \underline{\underline{\Theta(n)}}$

general term:

$$T(n) = T(n/3^k) + \frac{n}{3^0} + \frac{n}{3^1} + \dots + \frac{n}{3^{k-1}}$$

$$= T(n/3^k) + n \left[\frac{1}{3^0} + \frac{1}{3^1} + \dots + \frac{1}{3^{k-1}} \right]$$

$r < 1$

$$\begin{aligned} a &= 1 \\ r &= 1/3 \\ n &= k \end{aligned}$$

GP

$$T(n) = T(n/3^k) + \frac{1 * \left(1 - \left(\frac{1}{3} \right)^k \right)}{1 - 1/3} = \frac{3 \left(1 - \frac{1}{3^k} \right)}{2}$$

Q48

$$T(n) = 2T(n/2) + \log n$$

76%

✓ A) $O(n)$

B) $O(n^2)$

C) $O(n \log n)$

D) $O(\log n)$

$$\left. \begin{array}{l} a=2 \\ b=2 \\ f(n)=\log n \end{array} \right\}$$

$$\log_b a = \log_2 2 = \underline{\underline{1}}$$

Case 1: $\log n = O(n^{1-\epsilon})$, some $\epsilon > 0$?

$\epsilon = 0.5$ ✓

True

$$\boxed{T(n) = O(n^1)} \Rightarrow \underline{\underline{O(n)}}$$

PYQ

given

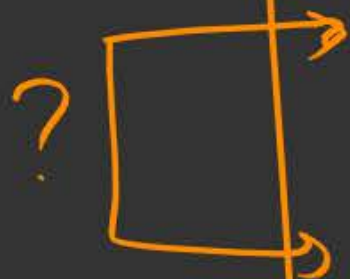
$$a, b \Rightarrow \omega(1)$$

$$T(n) = T(n^{1/a}) + 1$$

$$\Rightarrow \boxed{T(b) = 1}$$

$$\underline{\underline{T(n) = ?}}$$

Back-sub



A) $\Theta(\log_2 \log_2 n)$

B) $\Theta(\log_a \log_b n)$

C) $\Theta(\log_b \log_a n)$

D) $\Theta(\log_{a \times b} n)$

67.81.

$$n^{1/a^k} = b$$

$$\log(n^{1/a^k}) = \log(b)$$

$$\frac{1}{a^k} \log n = \log b$$

$$a^k = \frac{\log n}{\log b} = \log_b n$$

Take $\log()$

$$k \times \log a = \log(\log_b n)$$

$$k = \frac{\log(\log_b n)}{\log a}$$

$$k = \log_a(\log_b n)$$

$$T(n) = T(n'/a^k) + k$$

$$T(n) = T(1) + \log_a \log_b n$$

$$T(n) = 1 + \log_a \log_b n$$

$$T(n) = \Theta(\log_a \log_b n)$$

Special case: Masters Mtd

$$T(n) = T(\sqrt{n}) + 1 \longrightarrow \text{Special case}$$

$\Rightarrow$ change of variable mtd (std process)

$$T(n) = T(n^{1/2}) + 1 \text{ --- ①}$$

Let $n = 2^k$

$$\rightarrow T(2^k) = T(2^{k/2}) + 1 \text{ --- ②}$$

$$\left\{ \begin{array}{l} \text{Let } S(k) = T(2^k) \\ S(k/2) = T(2^{k/2}) \end{array} \right\}$$

$$T(2^k) = T(2^{k/2}) + 1$$

$$S(k) = S(k/2) + 1$$

$$\begin{array}{l} a=1 \\ b=2 \\ f(k)=1 \end{array}$$

$$\log_b a = \log_2 1 = \underline{\underline{0}}$$

Case 1:- $1 = O(k^{0-\varepsilon})$, some $\varepsilon > 0$?

Case 2:- $1 = O(k^0 * (\log k)^d)$ $\rightarrow$ Fails

a) $d > 0$?

$d = 0$

$$1 = O(\log k^0)$$

$$1 = O(1) \checkmark$$

$d = 0$

$$S(k) = O(k^0 * (\log k)^{d+1})$$

$$S(k) = O((\log k)^{d+1})$$

$$S(k) = O(\log k)$$

$$S(K) = O(\log K)$$

$$T(2^K) = O(\log K)$$

$$T(n) = O(\log K)$$



$$T(n) = O(\log \log_2 n)$$



$$2^K = n$$

$$K = \log_2 n$$

LIM $\rightarrow$ Long Integer Multiplication

'int a, b $\Rightarrow$ a+b $\rightarrow$ O(1)
 $\Rightarrow$ a*b $\Rightarrow$ O(1)

$a = 3429167832593$

$b = 73291352139867$

}

$\boxed{a \times b} \rightarrow \underline{\underline{lim}}$

$a[n] = \boxed{3 \mid 4 \mid 2 \mid 9 \mid \dots \mid \dots}$

$b[n] = \boxed{7 \mid 3 \mid 2 \mid 9 \mid 1 \mid \dots}$

$\begin{array}{r} 23 \\ \times 42 \\ \hline \end{array}$

$a+b$

for $i \rightarrow n$

$x_i = a_i + b_i$

$\rightarrow O(n)$

for $i \rightarrow n$

$j \rightarrow n$

$\rightarrow O(n^2)$

DnC Idea for LIM:

$$a = 4325 \rightarrow n$$



$$43 \times 10^2 + 25$$



$$a \times b = \underbrace{(43 \times 76)}_{n/2} \times 10^4 \dots$$

$$b = 7693 \rightarrow n$$



$$76 \times 10^2 + 93$$

Simple DnC

$$\begin{aligned} T(n) &= 4T(n/2) + bn, n > 1 \\ &= c, n = 1 \end{aligned}$$

mul

addn



Topic : Divide & Conquer

$$T(n) = 4T(n/2) + bn, n > 1$$
$$= C, n = 1$$

~~Adding~~ → TC = $\theta(n^2)$ → L/M → Recurrence

$$T(n) = aT(n/b) + f(n)$$

$$a = 4, b = 2, f(n) = bn \rightarrow \text{valid}$$

Case I: is $f(n) = O(n^{\log b^a - \epsilon})$, same $\epsilon > 0$

$$b_n = O(n^{2 - \epsilon}), \epsilon > 0 ? \rightarrow \text{Yes}$$

→ Satisfied



Topic : Divide & Conquer

~~$T(n) = \theta(n \log b^a) = \theta(n^2)$~~

~~Meet mule~~ matrix mul

(1) Non DnC $\rightarrow TC = \theta(n^3)$

(2) Simple DnC $\rightarrow TC = \theta(n^3)$

(3) Strassen's DnC $\rightarrow TC = \theta(n^{2.81})$

Similarly:

for L/M DnC $\rightarrow$ AK Optimization

$T(n) = 8T(n/2) + O(n^2)$

$T(n) = 7T(n/2) + O(n^2)$



Topic : Divide & Conquer

For AK optimization

$$\begin{aligned} T(n) &= 3T(n/2) + bn, n > 1 \\ &= C, n = 1 \end{aligned}$$

$$a = 3, b = 2, f(n) = bn$$

Case 1:

$$bn = O(n^{\log_2 3 - \epsilon}), \epsilon > 0$$

→ satisfied

yes ✓

$$T(n) = \theta(n^{1.58})$$

$$n^3 \rightarrow n^{2.81}$$

$$n^2 \rightarrow n^{1.58}$$

$$\log_2 3 = \underline{\underline{1.58}}$$

$$\underline{\underline{n^2 \rightarrow n^{1.58}}}$$



Topic : Divide & Conquer

Discussion so far:-

- (1) Non-DnC $\rightarrow$ TC = $\theta(n^2)$ $\rightarrow$ 2 loops
- (2) Simple-DnC $\rightarrow$ TC = $\theta(n^2)$
- (3) AK Optimization $\rightarrow$ TC = $\theta(n^{1.58})$

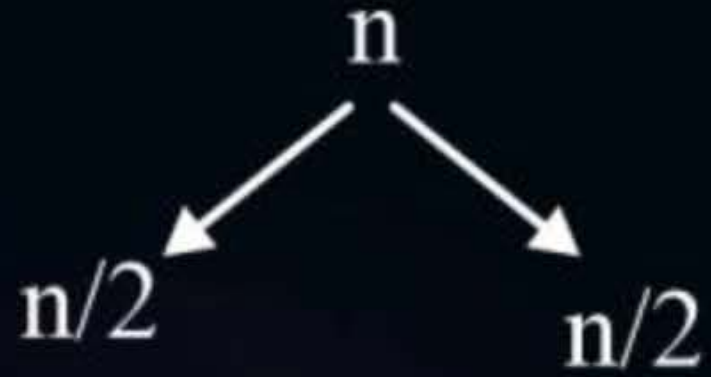


Topic : Toom & cook optimization

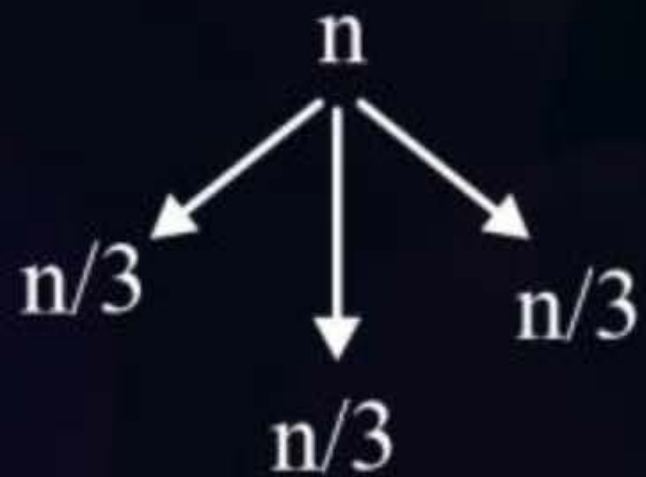


Idea:-

2-way split



3-way split



Topic : Recurrence after 3-way split

(1) Simple DnC $\Rightarrow T(n) = 2T(n/3) + n$

$$n^{\log_3 2} = n^2$$

(2) AK optimization DnC $\Rightarrow T(n) = \underline{8T(n/3)} + n$

$$n^{\log_3 8} = n^{1.89}$$

(3) T & C Optimization DnC $\Rightarrow \underline{5T(n/3)} + n$

$$n^{1.46}$$

$$\log_3 5 = \underline{1.46}$$



Topic : Summary to use in GATE

Summary to use in GATE

- (1) 2-way split: $DnC \rightarrow 4$, $AK \rightarrow 3$, $T \ \& \ C \rightarrow \underline{3}$
- (2) 3-way split: $DnC \rightarrow \underline{9}$, $AK \rightarrow \underline{8}$, $T \ \& \ C \rightarrow \underline{5}$
- (3) 4-way split: $DnC \rightarrow \underline{16}$, $AK \rightarrow \underline{15}$, $T \ \& \ C \rightarrow \underline{\underline{7}}$

Recurrence equation,

- (1) $DnC \rightarrow 16T(n/4) + bn, n > 1$
- (2) $AK \text{ opti.} \rightarrow 15T(n/4) + bn, n > 1$
- (3) $T\&C \text{ opt.} \rightarrow 7T(n/4) + bn, n > 1$



Topic : Summary to use in GATE

Generalised for k-way split

V. imp

$$\text{DnC} \rightarrow \underline{k^2}$$

$$A_k \rightarrow \underline{(k^2 - 1)}$$

$$T\&C \rightarrow \underline{(2^k - 1)} \quad (2k - 1)$$

The generalized time complexity recurrences for k-way split:

(1) $\text{DnC} \rightarrow T(n) = \underline{k^2} T(n/k) + bn$

(2) $\underline{A_k \text{ opti.}} \rightarrow T(n) = \underline{(k^2 - 1)} T(n/k) + bn$

(3) $\underline{T\&C \text{ opt.}} \rightarrow T(n) = \underline{(2k - 1)} T(n/k) + bn$



2 mins Summary



Topic

Topic

Topic

Topic

Topic

LIM
DnCV ✓



THANK - YOU